ASIAN INSTITUTE OF MANAGEMENT

SCHOOL OF EXECUTIVE EDUCATION AND LIFELONG LEARNING

Artificial Intelligence and Machine Learning

Module 1: Overview of AI and ML

M1 – Summary – Index

Introduction to AI and ML

- Emphasises the importance of harnessing data to redefine market boundaries
- Introduces AI and ML as pivotal technologies in the digital era
- Covers the agenda: basics of AI and ML, real-world application and industry trends
- Inspirational examples showcase AI's role in transforming industries like healthcare, retail and manufacturing

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Data Science as Art and Science

- Data Science combines creativity and analytics, likened to painting a masterpiece with algorithms
- Stresses the importance of creating models and insights that are both accurate and actionable
- Highlights the integration of AI, ML and Deep Learning (DL) to uncover complex patterns and insights

Real-World Applications of AI and ML

- Waze: AI for real-time traffic prediction and route optimisation
- Gmail: AI-driven Smart Compose and Smart Reply features
- Netflix: Machine Learning to personalise content recommendations
- Facebook: Advanced facial recognition for photo tagging and security
- Tesla Autopilot: Deep Learning for self-driving vehicles
- Amazon Alexa: Voice recognition and response powered by DL

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Dynamic Pricing and Forecasting

- Use of ML to predict demand and optimise pricing in real-time
- Example of an e-commerce giant leveraging 158 trillion combinations to enhance Black Friday sales

AI Applications in Finance

Algorithmic Trading

- High-speed trading decisions based on data patterns

Fraud Detection

- AI identifies anomalies to prevent fraudulent activities

Types of AI

Artificial Narrow Intelligence (ANI)

- Specialised AI performing specific tasks efficiently, like chatbots, voice assistants and recommendation systems
- Example: Healthcare diagnostic tools detecting diseases with high precision

Artificial General Intelligence (AGI)

- Future concept of AI capable of multi-domain learning, understanding, and reasoning like human intelligence
- Example: Autonomous robots adapting to diverse unstructured environments

Artificial Superintelligence (ASI)

- A theoretical stage where AI surpasses human intelligence in all domains, including creativity and decision-making
- Raises ethical and safety concerns about its societal impact

Machine Learning

ML is the science and art of enabling computers to learn from data and improve their performance without explicit programming. Introduced by Arthur Samuel (1959) and refined by Tom Mitchell (1997), ML involves learning from experience to improve performance on specific tasks.

Transition From Rules to ML for Spam Detection

Figure 1:Traditional Approach

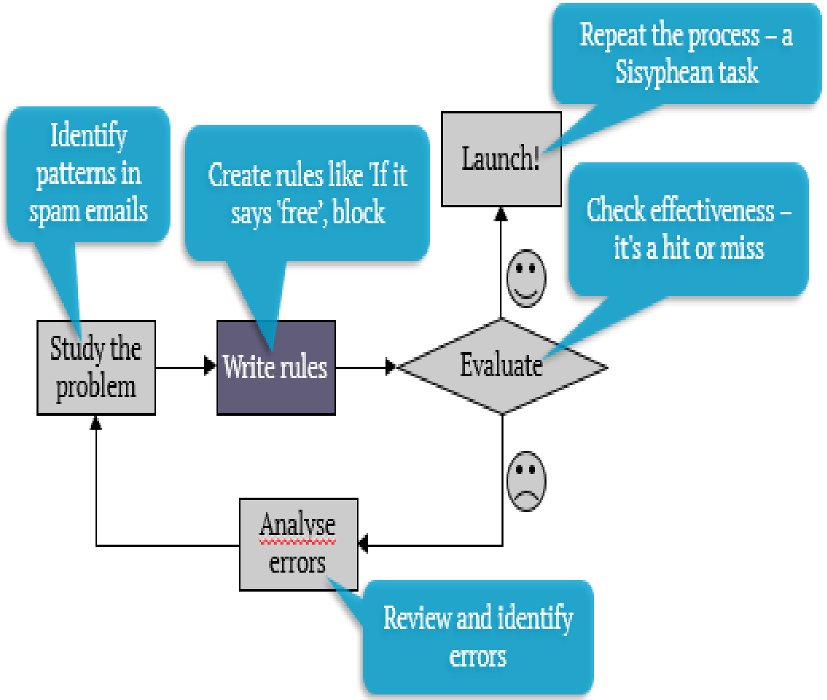
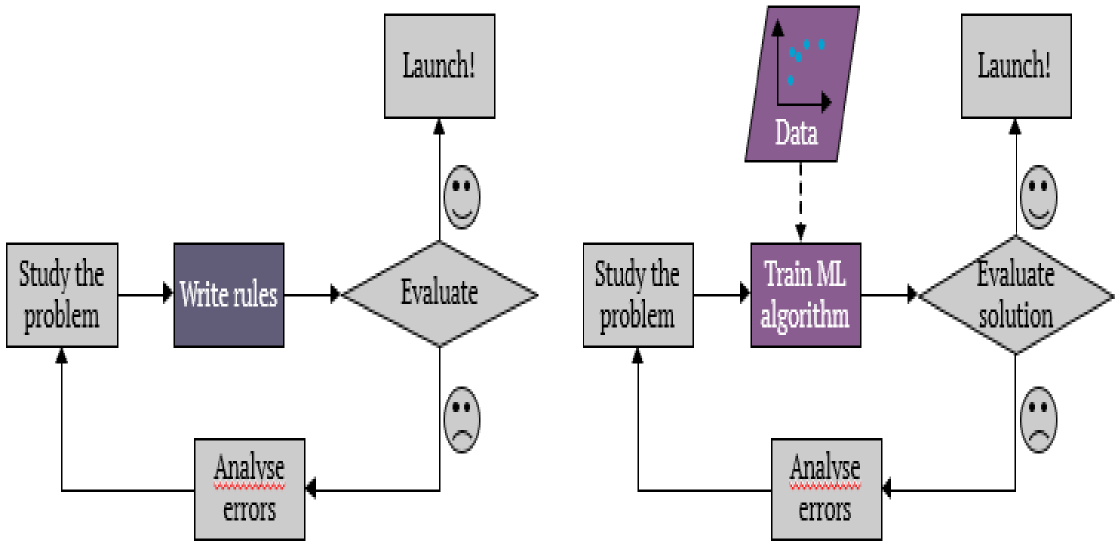


Figure 2: Machine Learning strides in



Transition From Rules to ML for Spam Detection

Figure 3:The new era

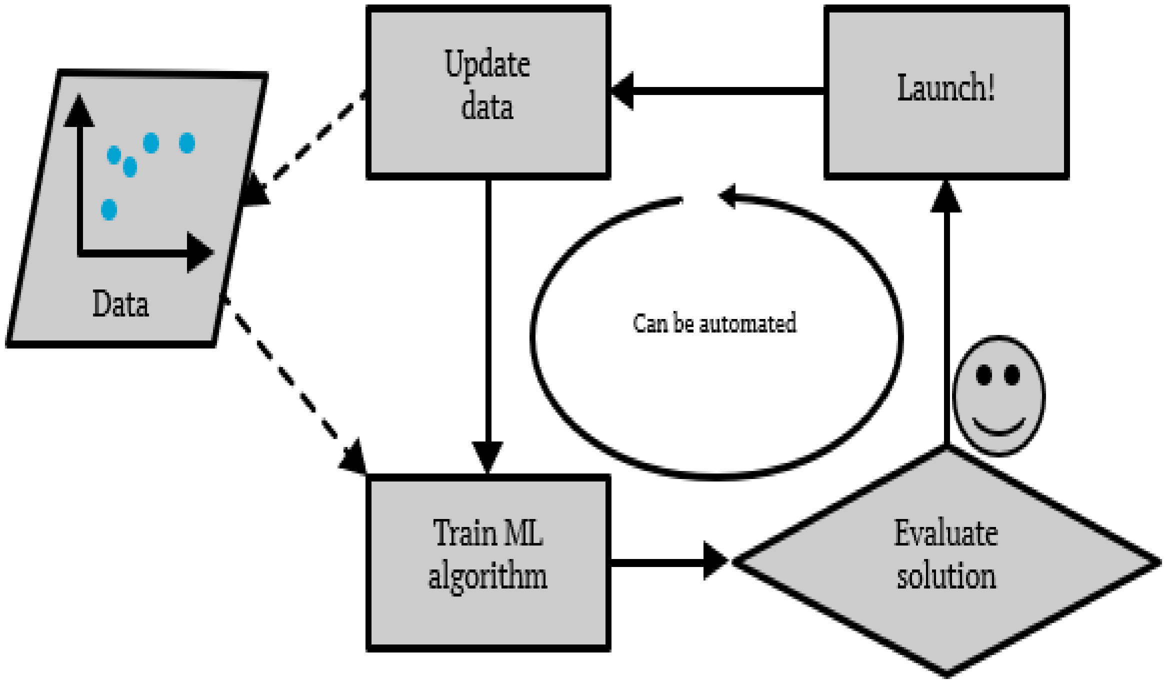
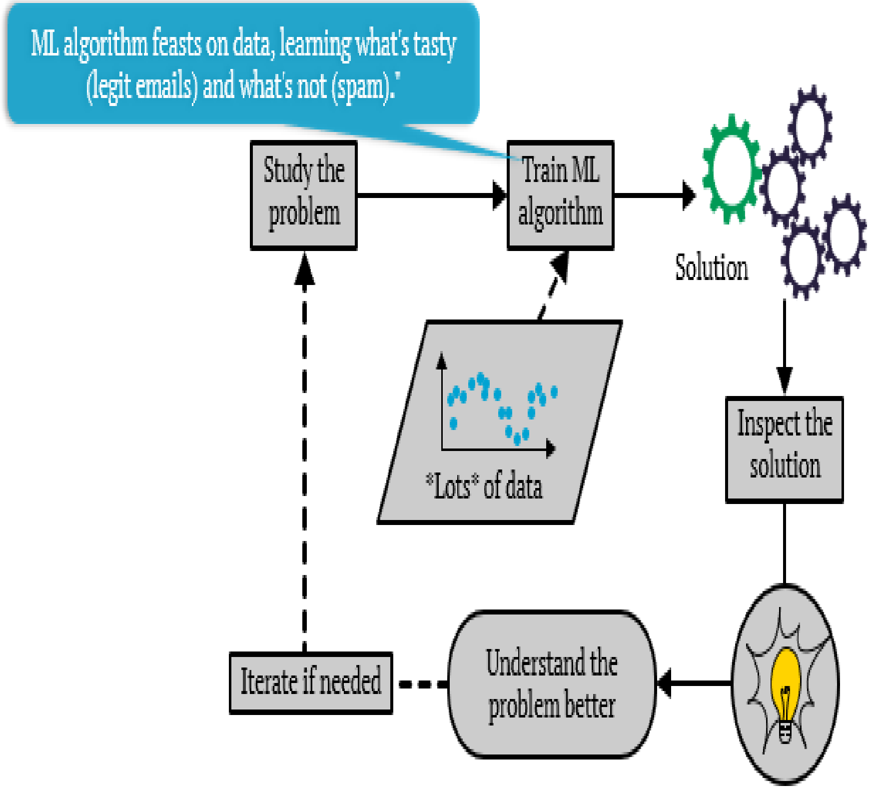


Figure 4:The cyclical genius of Machine Learning



Types of Machine Learning

Supervised Learning

- Supervised learning involves training a model on labelled data, where each data point is paired with a corresponding label. The goal is to learn a mapping between inputs (features) and outputs (labels)
- Used for tasks like classification (categorising data into predefined groups) and regression (predicting continuous numerical values)

Unsupervised Learning

- unsupervised learning works on unlabelled data to discover hidden patterns or structures within the dataset
- Used for tasks like clustering (grouping similar data points together) and dimensionality reduction (simplifying data while retaining important information)

Semi-Supervised Learning

- Combines a small amount of labelled data with a large amount of unlabelled data to improve learning performance
- Ideal for scenarios where obtaining labelled data is expensive or time-consuming but a large volume of unlabelled data is available

Reinforcement Learning (RL)

- Reinforcement learning, an agent learns by interacting with an environment and receiving feedback in the form of rewards or penalties for its actions
- RL focuses on optimizing long-term outcomes through trial and error, making it suitable for dynamic and complex decision-making tasks

ML Applications by Type

Supervised Learning

- Linear regression for predicting stock prices; logistic regression for hiring

Semi-Supervised Learning

- Document classification using limited labelled data with a large set of unlabelled data; medical image analysis for disease detection

Unsupervised Learning

- Customer segmentation and anomaly detection in fraud prevention

Reinforcement Learning (RL)

- Highlighted through AlphaGo's strategy improvement

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Challenges in Machine Learning

Insufficient Data

Limits model accuracy and decision-making

Data Bias and Representation

Ensures data diversity for accurate insights

Underfitting

Simplistic models failing to capture data complexity

Overfitting

Models overly tuned to training data, leading to poor generalisation

Data Quality and Feature Engineering

- Cleaning and curating datasets to enhance model performance
- Feature selection and extraction to identify meaningful variables

Validation and Model Evaluation

- Techniques such as test sets, cross-validation and hyperparameter tuning to ensure robustness
- Importance of generalisation error in measuring real-world applicability

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Applications of AI and ML in Different Fields

Natural Language Processing (NLP)

- Enables applications like translation, speech transcription, virtual assistants, sentiment analysis, and content moderation

Computer Vision

- Extracts labels and attributes from images (e.g., objects, gender, OCR for text)
- Enables technologies like facial recognition, video streaming analysis, and automated content moderation

Predictive Analytics

Embedded in dashboards for real-time event detection, decision support and risk modelling

User Experience Enhancement

- Combines text mining and NLP for better customer engagement
- Examples include chatbots and conversational platforms, which now integrate speech, gestures and other interfaces

Process Automation

- AI streamlines tasks in smart buildings, energy grids, supply chains and warehouses
- Robots and automation improve operations efficiency

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Case Studies of AI in Action

Sales Volume Forecasting

- AI and ML replaced traditional forecasting methods, leading to improved decision-making and operational savings of $2.5 million annually
- Steps included cleaning and filtering data, creating product-region pairs and feeding pre-processed weekly sales data into ML models

Customer and Market Segmentation

An online retailer used AI to segment markets by demographics and behaviour, leading to a 30% increase in engagement and a 20% rise in sales.

Biometric Authentication

- A bank implemented facial recognition and fingerprint scanning for seamless customer authentication
- Results included a 40% decrease in account takeovers and improved customer satisfaction

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Ethical Challenges of AI

Gender Bias

- AI algorithms often reflect the biases present in their training data
- This can lead to discriminatory outcomes. For instance, search engines might deliver biased results that reinforce gender stereotypes, perpetuating societal prejudices

Lack of Transparency

- AI decisions are not always easy to understand, posing challenges to accountability
- The lack of transparency makes it difficult to interpret how AI systems arrive at specific conclusions or recommendations

Discrimination and Fairness

- AI decisions can unintentionally lead to unequal treatment
- Biases in data or algorithm design can result in discriminatory practices, disadvantaging certain groups

Privacy and Security Concerns

- As AI evolves, the need to protect data privacy and security becomes more critical
- Improper handling of sensitive data can lead to privacy breaches and misuse

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